

A quick look at the IoT landscape in India

Vision of DeITY

- To create an IoT industry in India with a turnover of US\$ 15 billion by 2020 (around five per cent to six per cent of the global IoT industry)
- To develop IoT products specific to Indian needs in all possible domains
- To have strong emphasis on R&D, demo centres, standards, governance and so on

Focus areas

- Smart technology to manage cities, water, transport, environment, health, agriculture, waste management, safety, supply chains and more

Investment from the government

- 100 smartcities to cost approximately ₹ 70.6 billion

Challenges need to be addressed

- Getting devices connected, given the Indian scenario of extreme temperatures, high levels of humidity and dust, lack of clean and continuous power, inadequate telecom coverage, among others, can be daunting
- Security and privacy: government policy regarding user privacy in IoT networks needs to be framed in order to prevent theft, tampering of devices and data
- IoT business models need to first address affordability to a billion people
- Technology and standards: An IoT solution will involve many stakeholders, so alignment to global standards is important

beyond consumer electronics, showing promise in other markets and applications including energy, water, traffic management, industrial IoT, surveillance and security.

High growth in this market is anticipated on account of ongoing developments in IoT technology, which provide better connectivity and coverage as well as real-time monitoring/tracking of services and systems across diverse industry verticals to reduce operational and manpower costs.

Various government projects such as smartcities, smart transportation and smartgrids, among others, are also expected to propel the adoption of IoT technology in the country. Therefore the initial smartproduct-oriented applications in white goods, cars, etc, will be extended to improve supply chain agility, manufacturing and operations efficiency and the like.

India advantage

Government policies will play a major role in accelerating early adoption of the IoT in the country. State-driven applications like smartcities are still considered as key factors for large-scale adoption. Prime Minister's vision of smartcities and a digital India will leverage the IoT to address and solve problems that large cities are facing in becoming smart.

Issues such as transportation systems, parking, lighting, waste management, water management and women's safety are matters of top priority. We will need sensors at strategic points to collect data, applications to analyse it and analytics to ensure quick decision-making.

Digital infrastructure is expected to create huge opportunities for technology companies across verticals and in key applications within smartcities, smart health systems

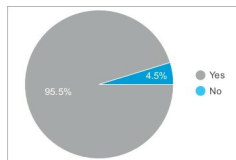


Fig. 1: Predicted growth of the Indian IoT industry in FY 2016-17

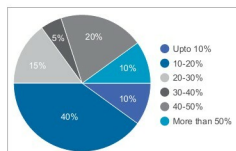


Fig. 2: Expected growth rate of the Indian IoT industry in FY 2016-17

and smart transportation. In India, technology visionaries and several startups are flooding the market with smartdevices, with the intent of taking the IoT adoption to the next level. The following domains would see heightened activity both in innovation and in adoption of the IoT:

- Urban safety management (including transportation)
- Environmental monitoring
- Delivery of government services
- Healthcare
- Utilities

Not only the government, even businesses can reap many benefits by using the IoT ecosystem. The biggest advantage is that many Indian industries have not been automated and the IoT offers an opportunity to leapfrog a couple of generations of manufacturing technology and go straight into the IoT era, saving a lot of time and money.

For example, manufacturers can remotely monitor the condition of equipment and look for indicators of imminent failures like vibration, temperature or pressure outside normal limits. This means that they have to make fewer site visits and employees can devote time on other critical issues, thereby reducing fixed and variable costs. For the customer, it means less disruption, increased uptime and, ultimately, higher satisfaction.

Taking the IoT to the next level, businesses can even offer pay-per-use options, inclusive of all hardware, installation and servicing. The IoT is already heralding transformation across industries and will bring about significant changes in the coming years. It can help organisations utilise their business infrastructure and assets in innovative ways to offer new services and deliver additional revenue. Moreover, deriving meaningful information from the huge volumes of data that the IoT produces can improve decision-making and enable proactive, predictive insights into work processes.